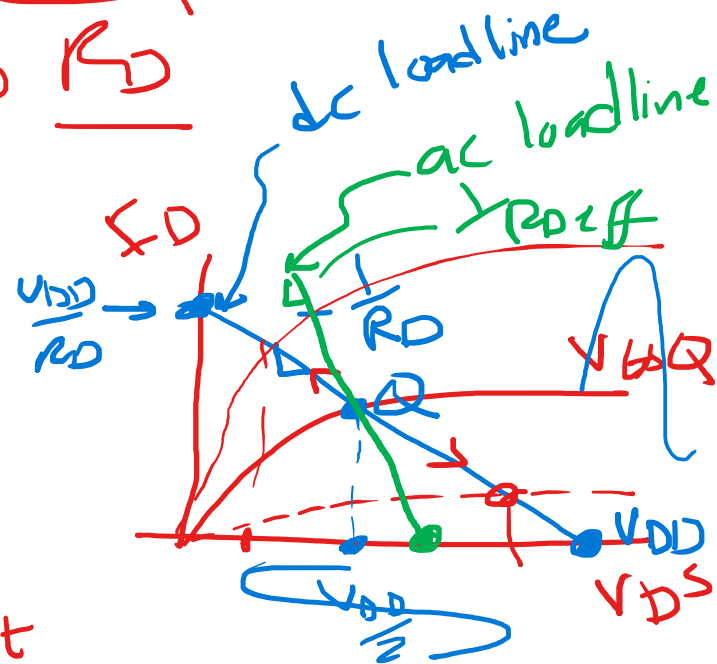
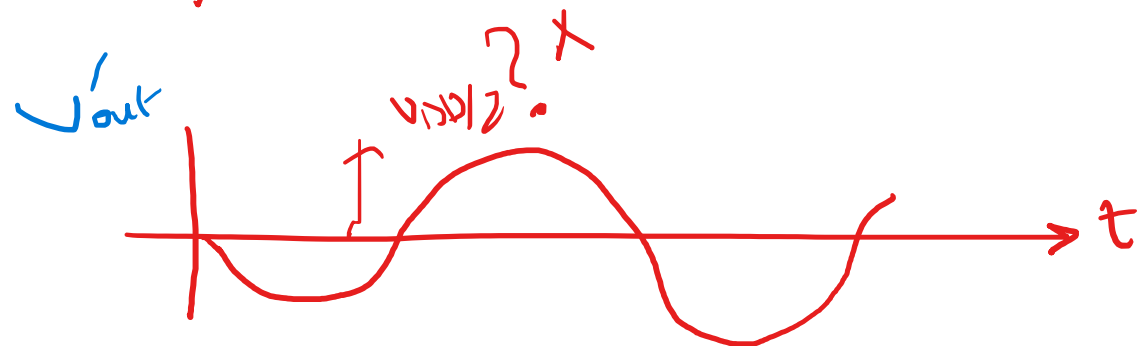
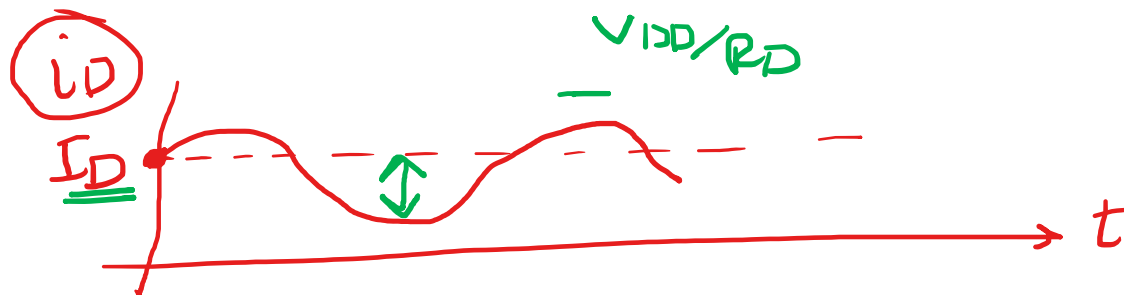
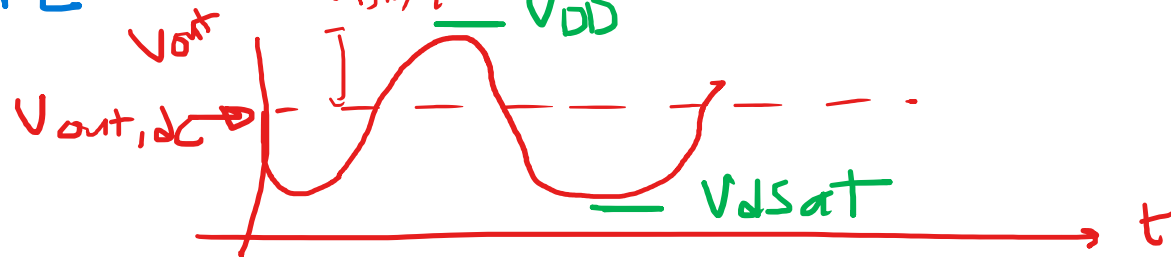
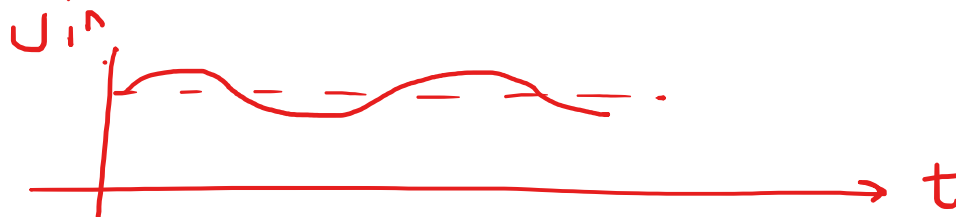
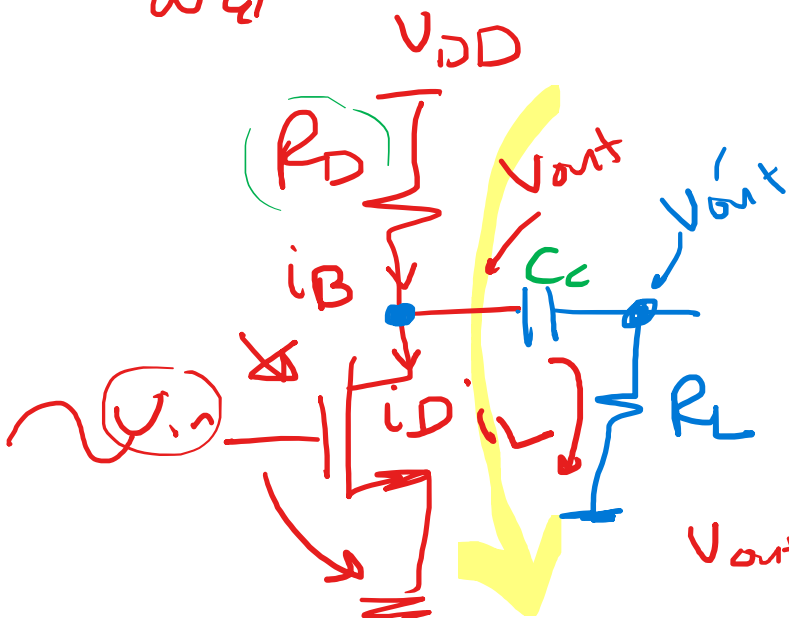


$$X_C = \frac{1}{\omega C} \downarrow \downarrow$$

$$ac: R_{o,eff} = R_D || R_L < R_D$$

$$V_{out,DC} = V_{DD} - \frac{I_D R_D}{V_{R_D}}$$



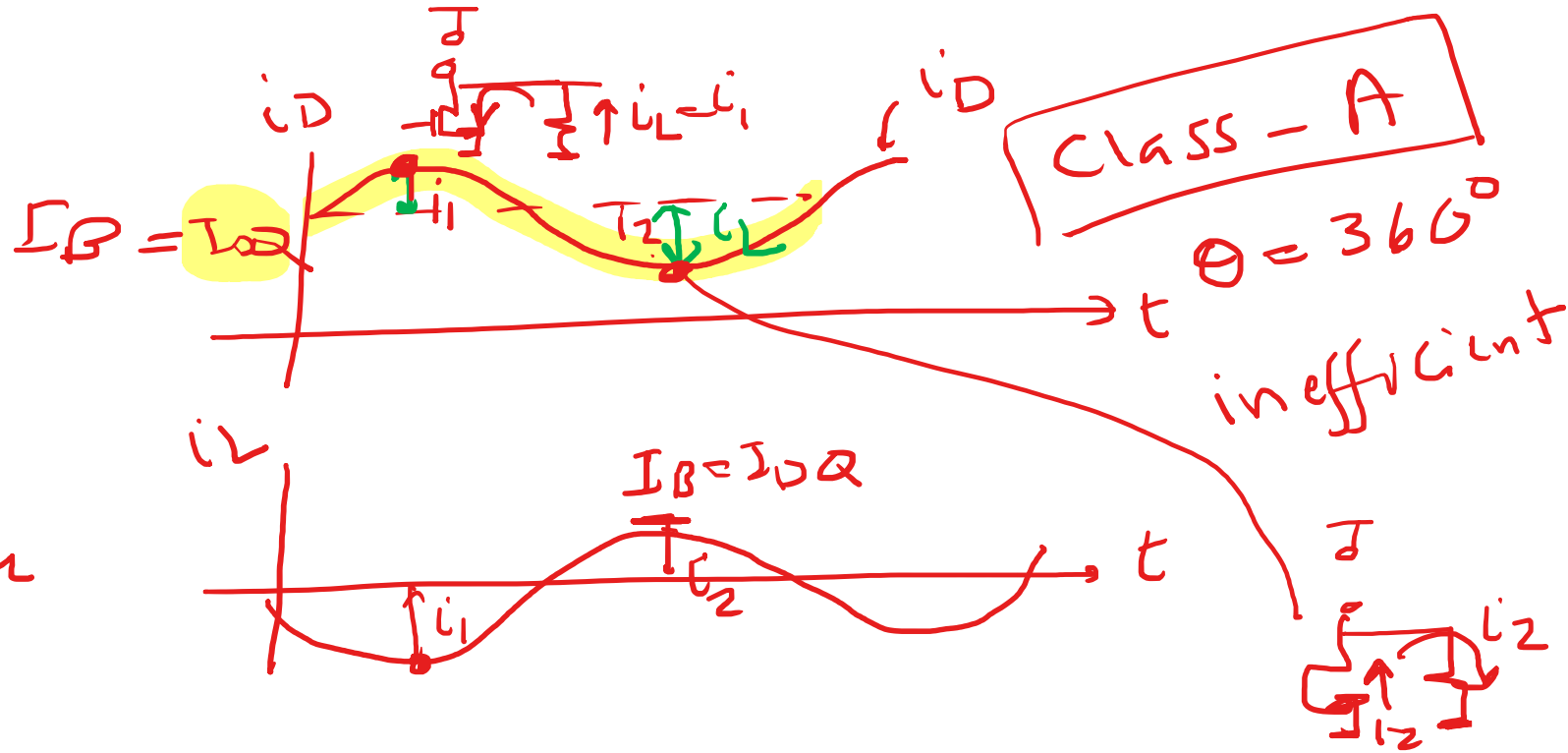
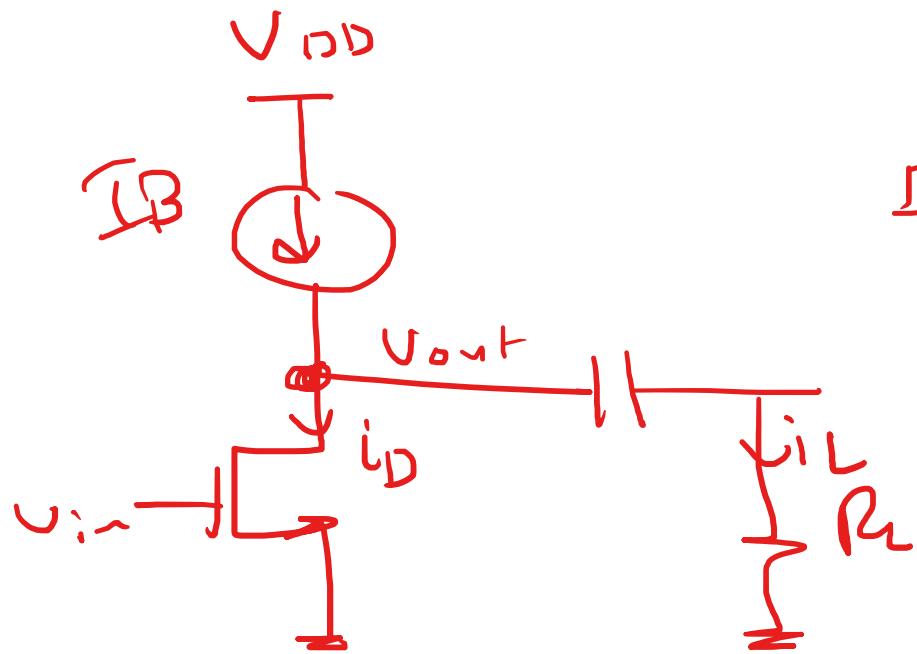
$$V_{DD} = \underbrace{I_D R_D}_y + \underbrace{V_{DS}}_x$$

$$I_D = -\frac{V_{DS}}{R_D} + \frac{V_{DD}}{R_D}$$

$$y = mx + c$$

$$i_B = i_L + i_D$$

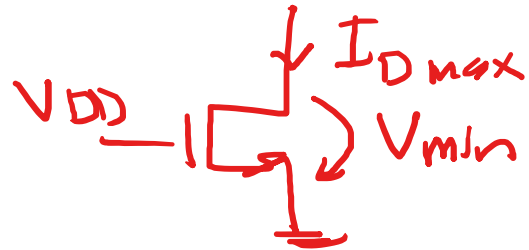
$$i_L = \underline{i_B} - (i_D)$$

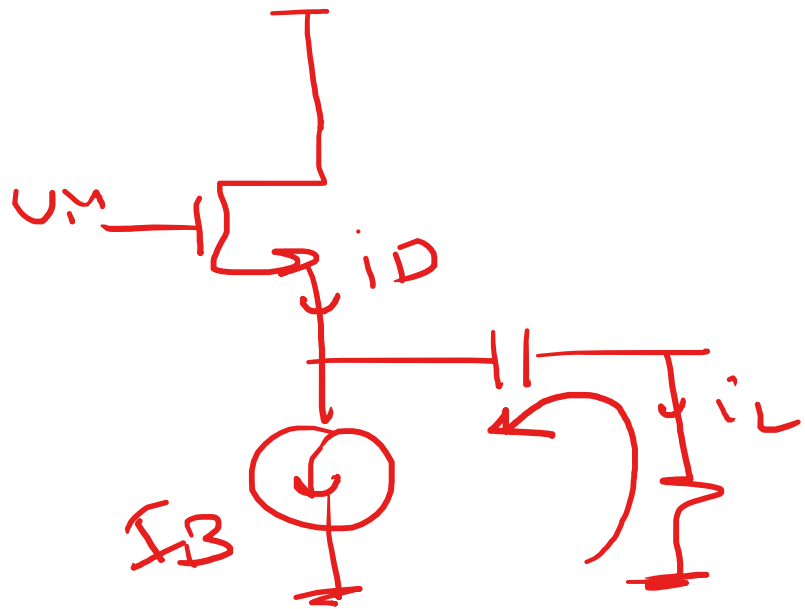


$$i_L = I_B - i_D$$

$$I_{source, max} = I_B = I_{DQ}$$

$$|I_{sink, max}| = I_{Dmax} - I_B$$





$$i_L = i_D - I_B$$

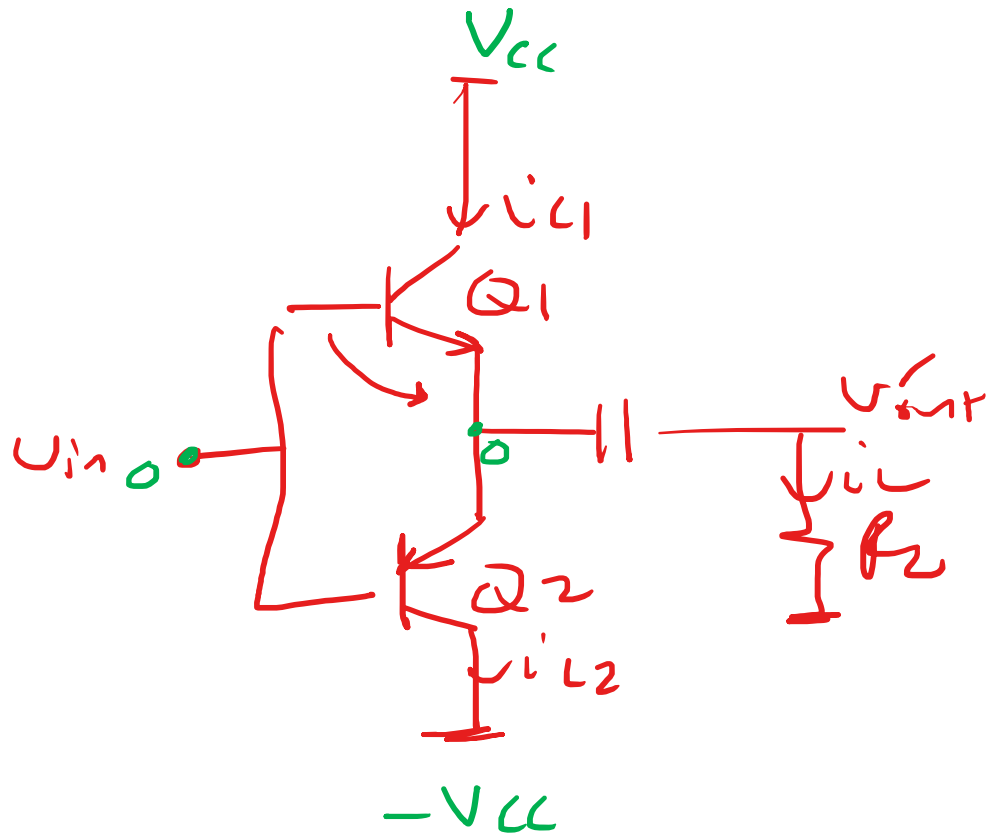
$$\begin{matrix} I_D \\ i_D \\ \textcircled{i_D} \end{matrix}$$

$$|I_{\text{sink, max}}| = |0 - I_B| = \textcircled{I_B}$$

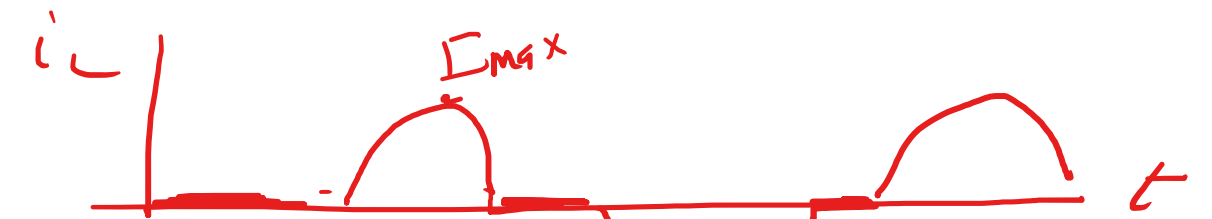
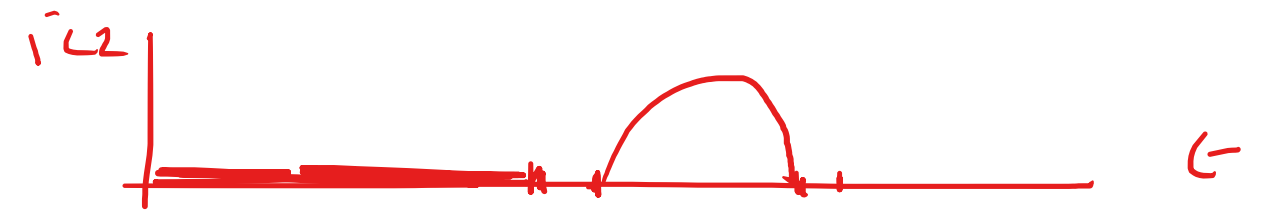
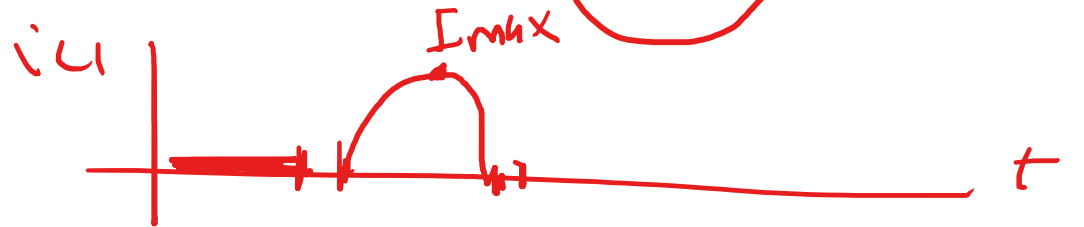
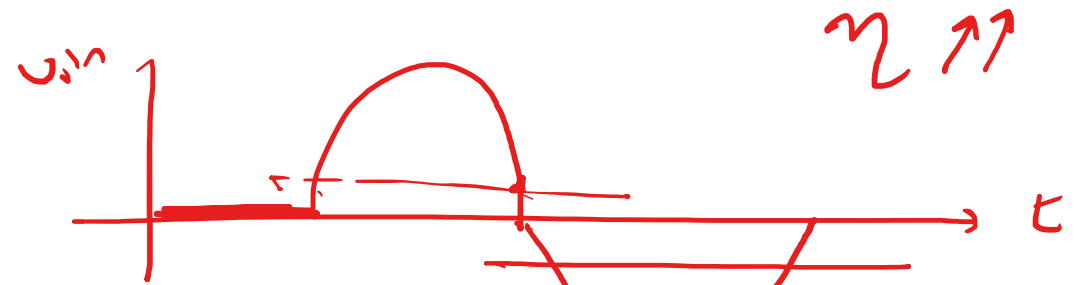
$$\textcircled{a} \quad v_{in, ac} = 0 \Rightarrow i_D = \textcircled{I_B} \quad \therefore$$

$$\eta = \frac{P_L}{P_{\text{total}}}$$

Class - B $\Theta \sim 180^\circ$

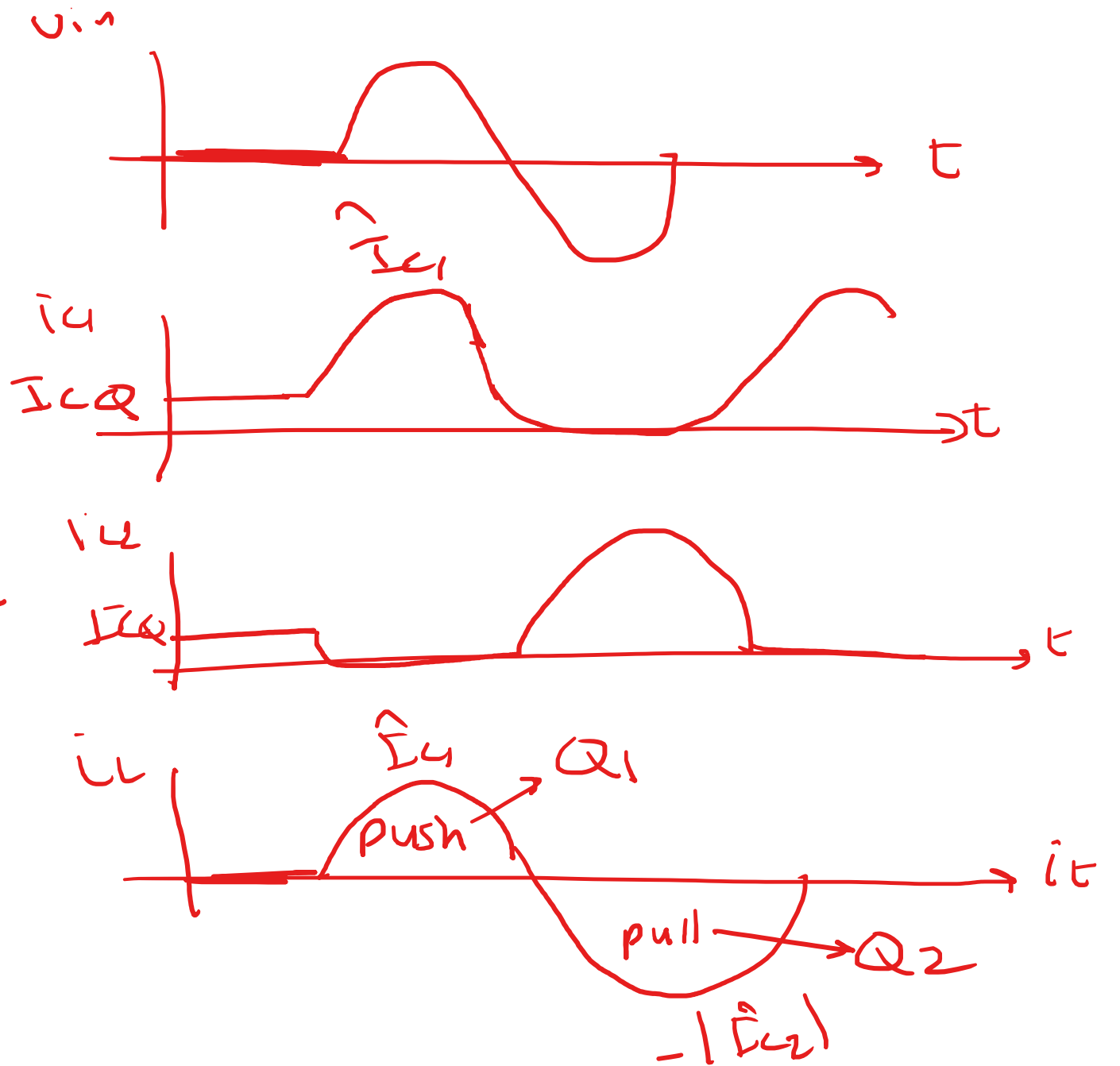
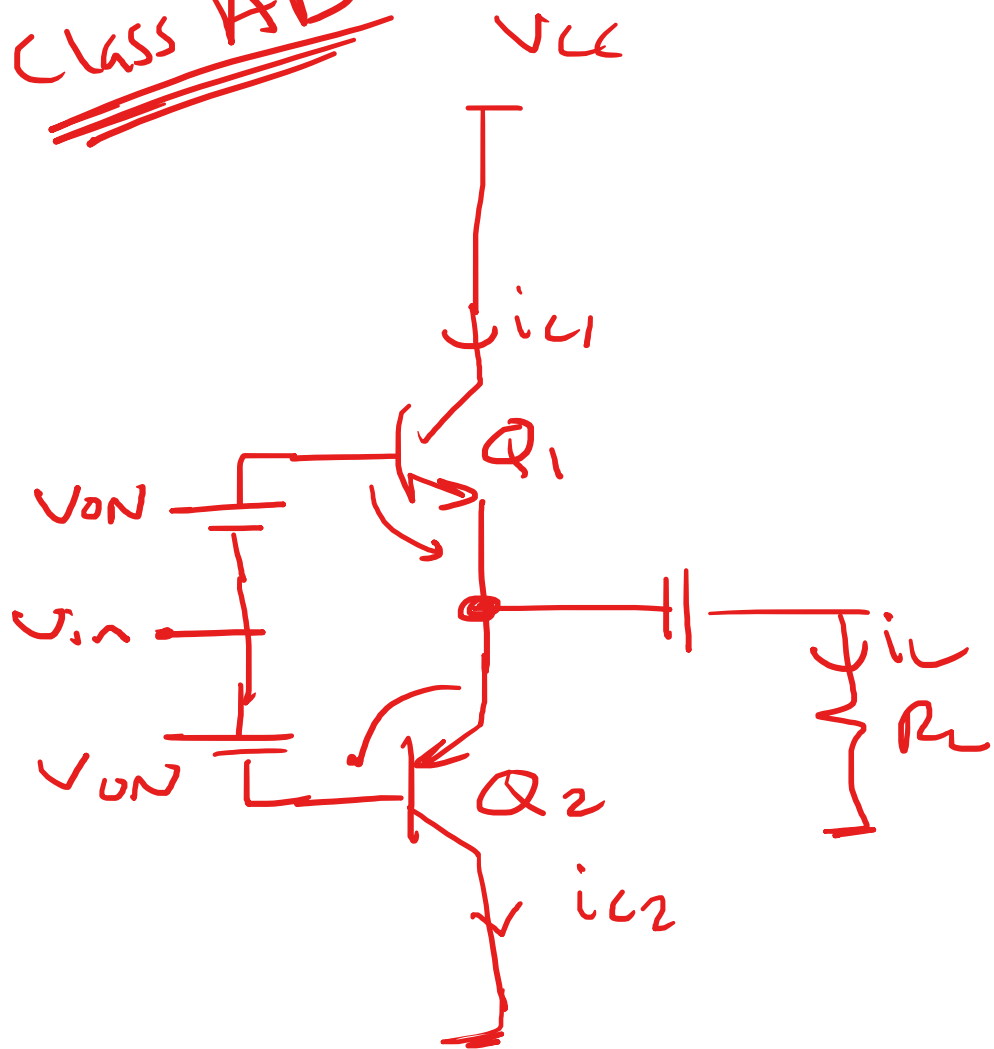


$$u_{out} = i_L \times R_L$$



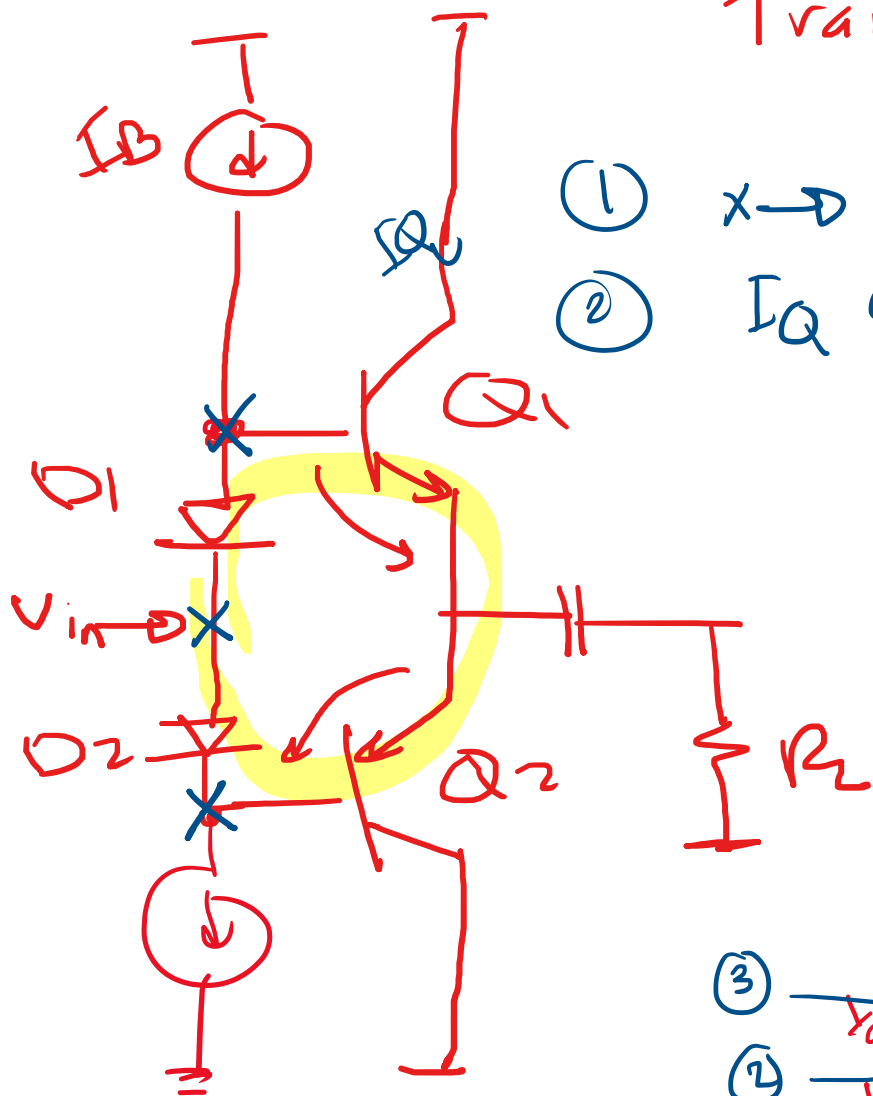
Cross-over distortion

CLASS AB



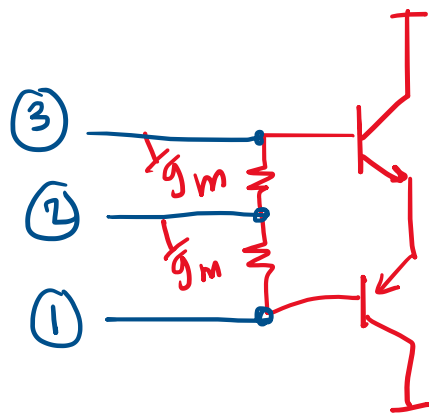
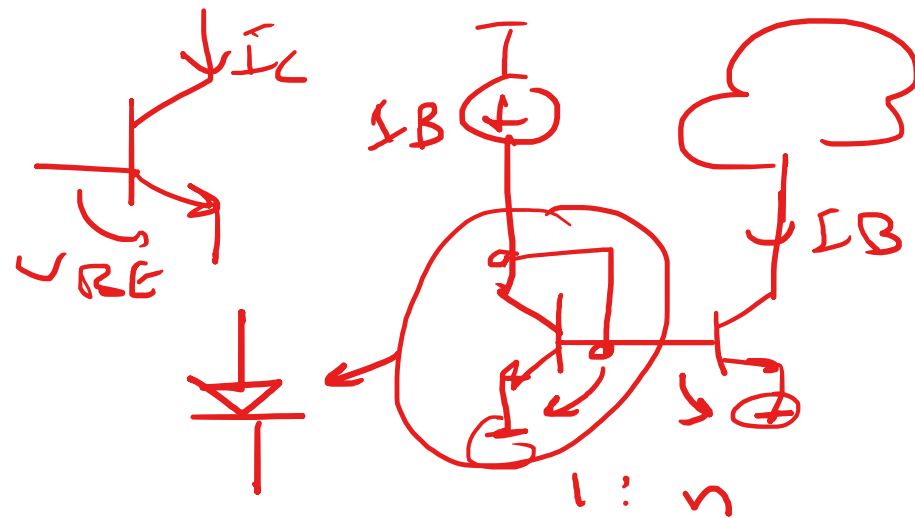
Translinear Loop

- ① $x \rightarrow HIN$
- ② I_Q Controlled

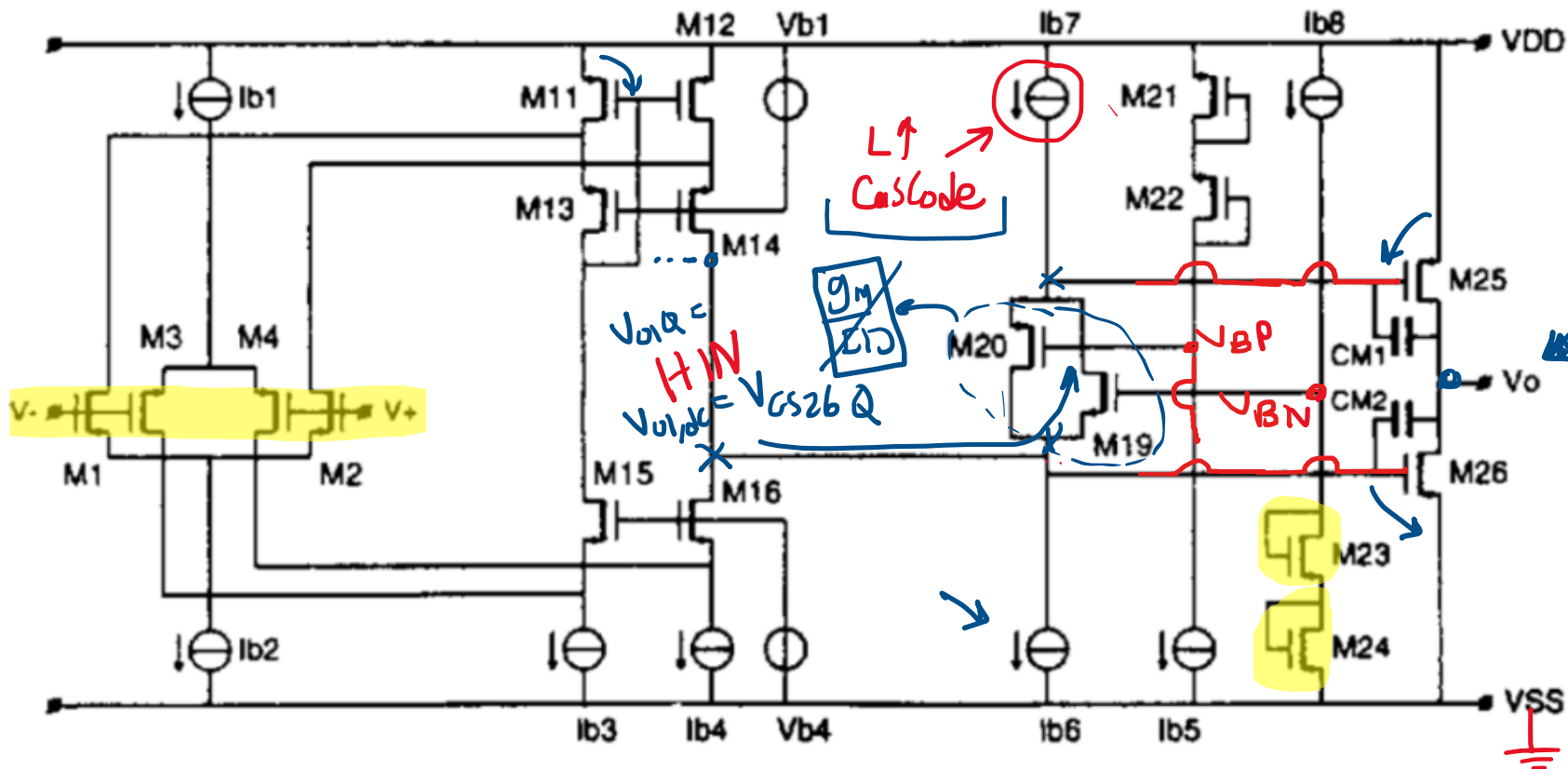


$$V_{D1} + V_{D2} = V_{BE1} + V_{BE2}$$

$$\textcircled{V_B} = V_{BE1} + V_{BE2}$$



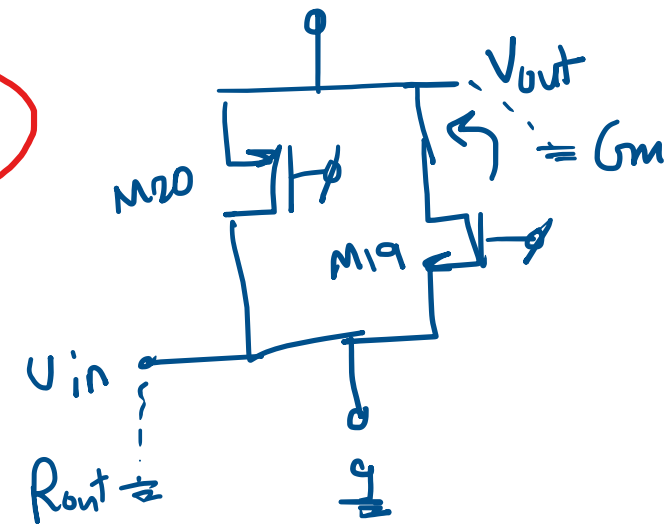
$$\frac{1}{g_m} \left(1 + \frac{R_D}{r_o} \right)$$



$$V_{B1} = V_{GS23} + V_{GS24} = V_{GS19} + V_{GS26}$$

$$G_m = g_{m19} + g_{ds20}$$

$$R_{out} = \frac{1}{g_{m20} + g_{ds19}}$$



Montecelli Biasing

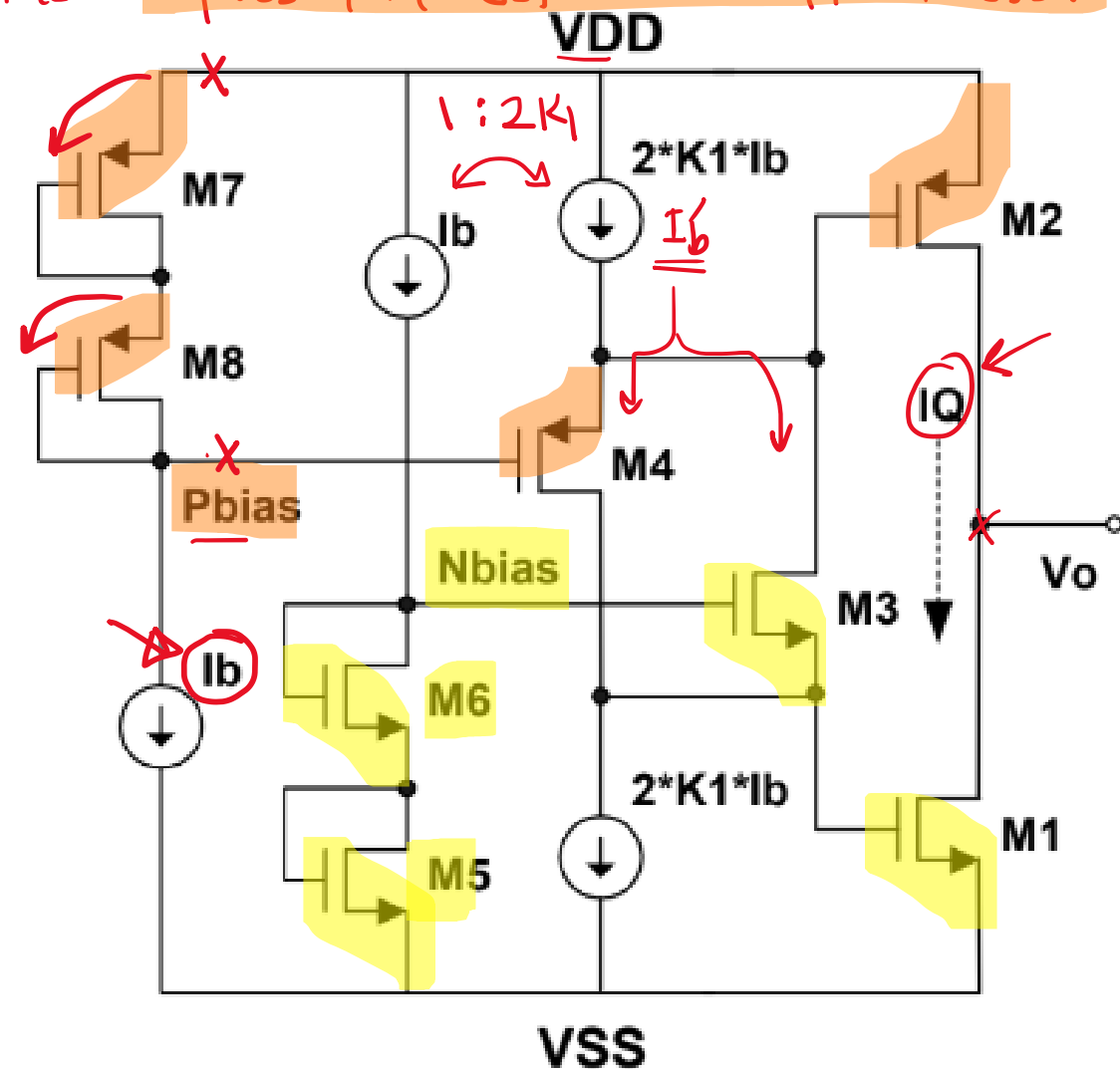
- Why does the current " $2 \cdot K_1 \cdot I_b$ " split equally between M3 and M4?
- If initially M4 has more current, how is this error corrected?

$$I_{D4} \uparrow \quad V_{GS4} \uparrow \quad V_{GS2} \downarrow \quad I_Q \downarrow$$

$$V_{GS1} \downarrow \quad V_{GS3} \uparrow \quad I_{D3} \uparrow \quad I_{D4} \downarrow$$

$$N_{bias} = V_{US6} + V_{US5} = V_{GS3} + V_{GS1}$$

$$V_{DD} - P_{bias} = \frac{|V_{GS2}| + |V_{GS1}|}{V_{DD}} = \frac{|V_{GS4}| + |V_{GS2}|}{V_{DD}}$$

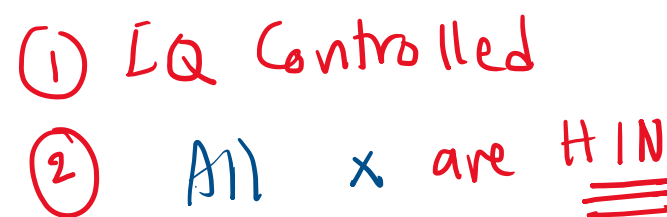


N bias = $V_{GS6} + V_{GS5} = V_{GS3} + V_{GS1}$ ←

-
- A hand-drawn diagram of a staircase with four steps, labeled 1, 2, 3, and 4 from left to right. The steps are drawn with red lines, showing the profile of the stairs. The steps are of varying heights and widths, with the fourth step being the tallest and the widest.

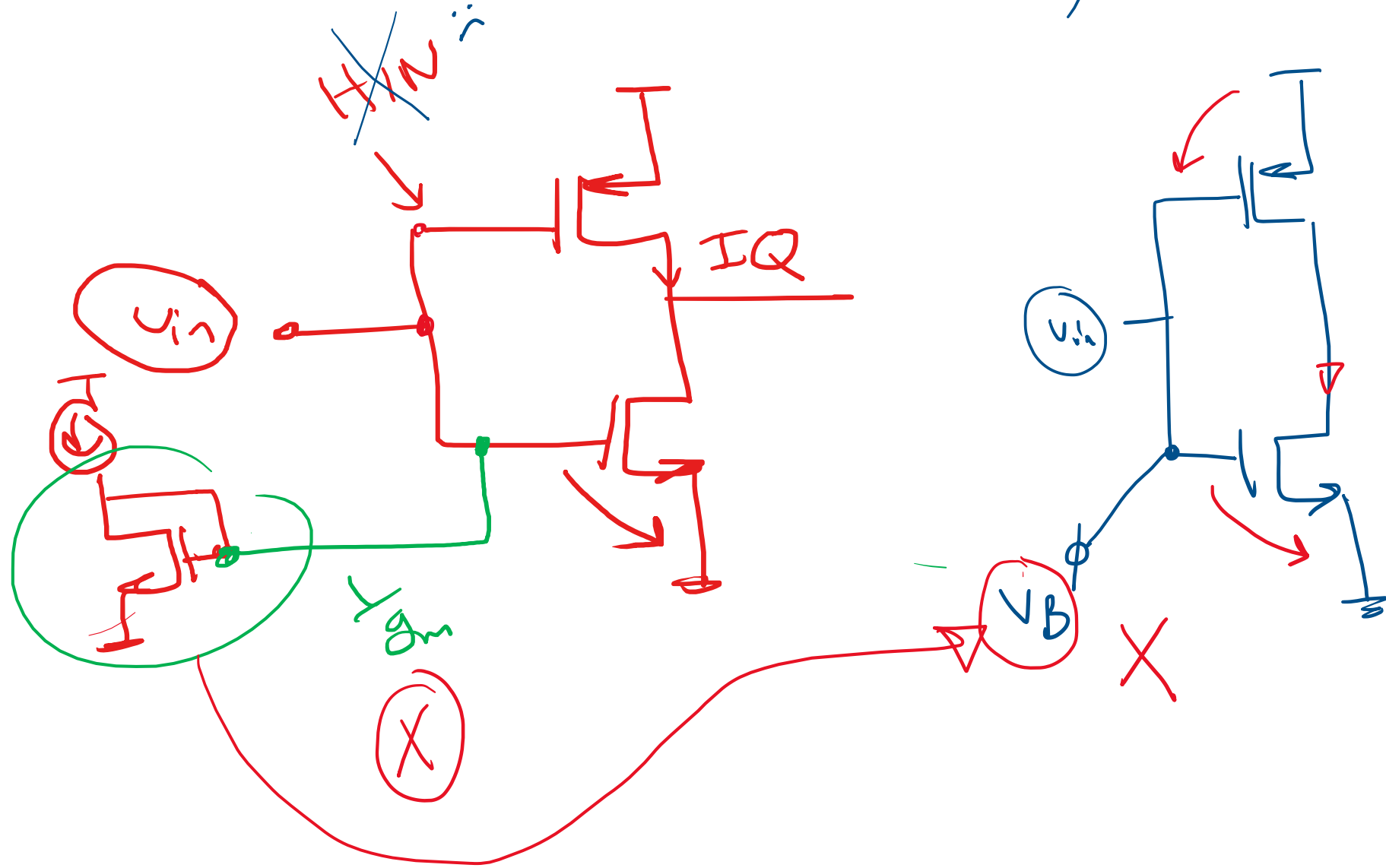
$10I_0$
 $\uparrow$
 $10I_0$
 $2I_0$
 $2I_0$
 I_0
 I_0
 I_0
 I_0

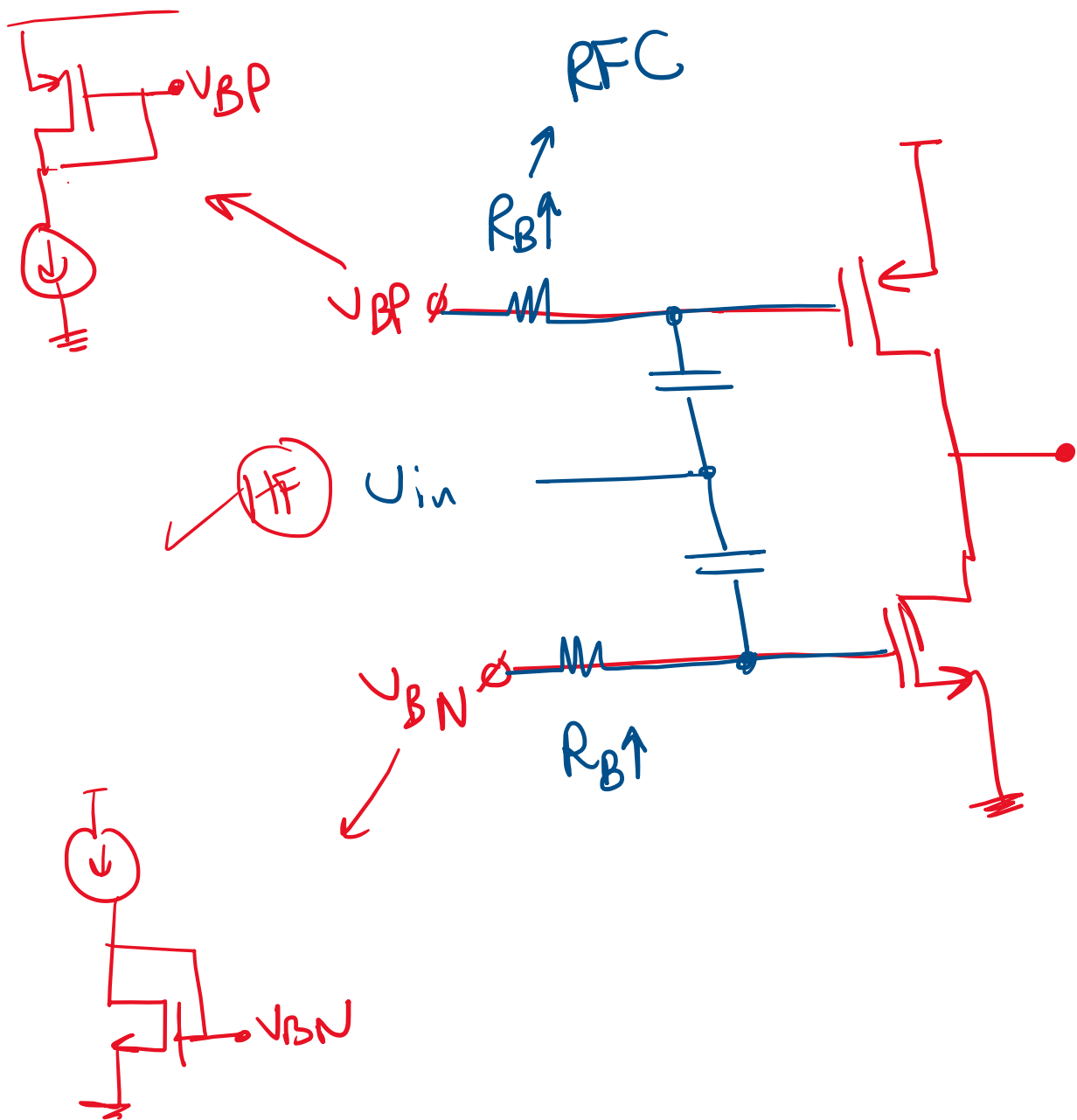
$\times 10$



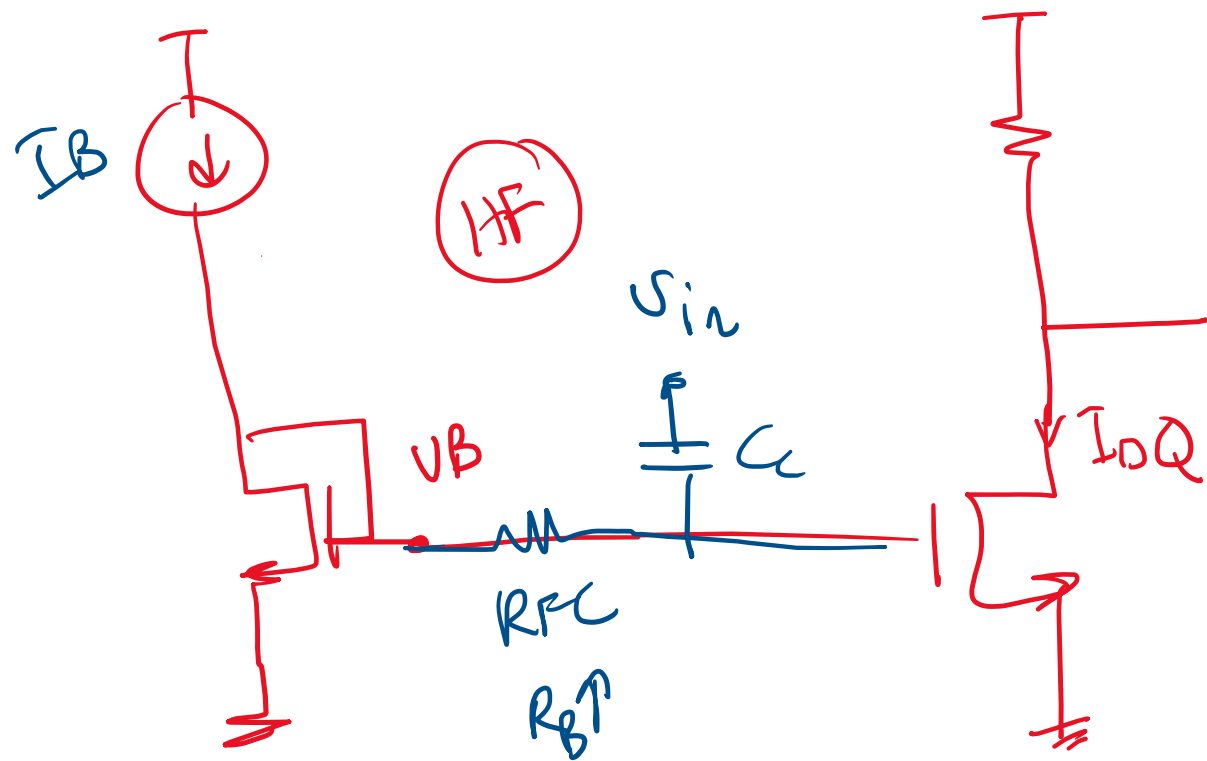
② All x are HIN

$$A_v = G_m R_{out} \quad \underline{\underline{H_{IN}}}$$





~~Baseband~~



~~Baseband~~